elegant-polimi-thesis documentation

This chapter is the result of the documentation generated from the source code. Since it's not strictly needed to use the template – most of the functions are straightforward – it's quite barebones.

This document contains them all, even those that are not exposed to the user, which are marked with "_" before.

All functions

User designated functions 1
- polimi-thesis 1
- polimi-article-format-thesis 3
- polimi-executive-summary 5
- _blank-toc 6
- frontmatter 6
- acknowledgements 6
- mainmatter 7
- appendix 7
- backmatter 7
- _lists 7
- _lists-entries-style 7
- nomenclature 8
- theorems-init 8
- subfigure 9
- toc 9
- list-of-figures 9
- list-of-tables 9
- theorem 9
- proposition 9
- lemma 10
- remark 10
- proof 10

Utilities 10
- _is-page-empty 10
- _empty-page 10
- _nameref 10
- _raggiera-image 10
- banner 11
- tab-numbering 11
- chapter-numbering 11
- header-number 12
- _sizes 12
- _document-state 12
- _document-type 12
- _localization 12
- bluepoli 12

Figures styling 12
- _subfigure-caption 12
- _style-figures 13

User designated functions

polimi-thesis

The main thesis formatting function.

Parameters

```
polimi-thesis(
  title: str,
  author: str,
  advisor: str,
  coadvisor: str arr,
  academic-year: str,
  tutor: str,
  cycle: str,
  chair: str,
  student-id: str,
  course: str,
  frontispiece: "phd" "deib-phd" "cs-eng-master" "classical-master",
  custom-logo: image,
  body
) -> content
```

title `str`

Title of the thesis.

Default: `"Thesis Title"`

author `str`

Author of the thesis.

Default: `"Name Surname"`

advisor `str`

Advisor of the thesis.

Default: `"Advisor"`

coadvisor `str` or `arr`

Coadvisor(s) of the thesis.

Default: `"Coadvisor"`

academic-year `str`

Academic year of the thesis. If empty, defaults to “2025-2026”.

Default: `""`

tutor `str`

Tutor of the thesis.

Default: `"Tutor"`

cycle `str`

Cycle of the thesis.

Default: `"XXV"`

chair `str`

Chair of the thesis.

Default: `"Chair"`

student-id `str`

Student ID.

Default: `"00000000"`

course `str`

Student course.

Default: `"Course Engineering"`

frontispiece `"phd"` or `"deib-phd"` or `"cs-eng-master"` or `"classical-master"`

Frontispiece of the thesis. Can be either: `phd`, `deib-phd`, `cs-eng-master` or `classical-master`.

Default: `"phd"`

custom-logo `image`

Custom logo of the thesis.

Default: `none`

polimi-article-format-thesis

The thesis article format styling function.

Parameters

```
polimi-article-format-thesis(
  title: str,
  author: str,
  advisor: str,
  coadvisor: str array,
  academic-year: str,
  student-id: str,
  course: str,
  abstract: content,
  keywords: str,
  logo: path,
  body
) -> content
```

title `str`

Title of the thesis.

Default: `"Thesis Title"`

author `str`

Author of the thesis.

Default: `"Name Surname"`

advisor `str`

Advisor of the thesis.

Default: `"Prof. Name Surname"`

coadvisor `str` or `array`

Coadvisor(s) of the thesis.

Default: `"Prof. Name Surname"`

academic-year `str`

Academic year of the thesis. If empty, defaults to “2025-2026”.

Default: `""`

student-id `str`

Student ID.

Default: `"00000000"`

course `str`

Student course.

Default: `"Xxxxxxxxxxxx Engineering - Ingegneria Xxxxxxxxxxxx"`

abstract `content`

Abstract.

Default: `[]`

keywords `str`

Keywords, that appear below the abstract (and in the PDF metadata).

Default: `"word, word, word"`

logo `path`

Logo of the thesis.

Default: `"img/logo_ingegneria.svg"`

polimi-executive-summary

The executive summary styling function.

Parameters

```
polimi-executive-summary(
  title: str,
  author: str,
  advisor: str,
  coadvisor: str array,
  academic-year: str,
  course: str,
  logo: path,
  body
) -> content
```

title `str`

Title of the thesis.

Default: `"Thesis Title"`

author `str`

Author of the thesis.

Default: `"Name Surname"`

advisor `str`

Advisor of the thesis.

Default: `"Prof. Name Surname"`

coadvisor `str` or `array`

Coadvisor(s) of the thesis.

Default: `"Prof. Name Surname"`

academic-year `str`

Academic year of the thesis. If empty, defaults to "2025-2026".

Default: `""`

course `str`

Student course.

Default: `"Xxxxxxxxxxxx Engineering - Ingegneria Xxxxxxxxxxxx"`

logo `path`

Logo of the thesis.

Default: `"img/logo_ingegneria.svg"`

_blank-toc

Helper function to manually add blank space above an outline entry (similar to LaTeX's `\addtocontents{toc}{\vspace{1em}}`).

Parameters

```
_blank-toc(space: length) -> content
```

space `length`

Vertical space to add before the next outline entry.

Default: `1em`

frontmatter

Frontmatter section. Similar to LaTeX's `\frontmatter`, it is meant to be only used in the thesis. It sets the page numbering to `"i"` and `numbering: none` for headings.

Parameters

```
frontmatter(body) -> content
```

acknowledgements

Acknowledgements section. It sets `numbering: none` for headings.

Parameters

```
acknowledgements(body) -> content
```

mainmatter

Mainmatter section. Similar to LaTeX's `\mainmatter`, it is meant to be only used in the thesis. It sets to page numbering to `"1"`, heading numbering to `"1.1" + h()` and resets the page counter.

Parameters

```
mainmatter(body) -> content
```

appendix

Appendix section. Similar to LaTeX's `\appendix`. It sets heading numbering to `"A.1"` and resets their counter.

Parameters

```
appendix(body) -> content
```

backmatter

Backmatter section. Similar to LaTeX's `\backmatter`, it is meant to be only used in the thesis. It sets heading numbering to `none`.

Parameters

```
backmatter(body) -> content
```

_lists

Lists figure to make the list of tables, list of figures to appear in the table of contents.

Parameters

```
_lists() -> content
```

_lists-entries-style

Internal helper function to create the custom lists of figures and table.

Parameters

```
_lists-entries-style(
  outline-entry: outline-entry,
  kind: function
) -> content
```

outline-entry `outline-entry`

Outline entry to edit.

kind `function`

The kind of the outline entry element (image or table)

nomenclature

Displays a simple nomenclature with keys and values.

```
#nomenclature((
    "Polimi": "Politecnico di Milano",
    "CdL": "Corso di Laurea",
    "CCS": "Consigli di Corsi di Studio",
    "CFU": "Crediti Formativi
Universitari",
))
```

Nomenclature

POLIMI	Politecnico di Milano
CDL	Corso di Laurea
CCS	Consigli di Corsi di Studio
CFU	Crediti Formativi Universitari

Parameters

```
nomenclature(
  dict: dictionary,
  indented: bool
) -> content
```

dict `dictionary`

Dictionary that hold keys and values.

indented `bool`

Whether to indent or not the nomenclature.

Default: `true`

theorems-init

Utility function to initialize the theorem environments[1]:

- theorem
- proposition
- lemma
- remark

Parameters

```
theorems-init(body) -> content
```

[1]Provided by great-theorems package.

subfigure

Creates a new subfigure with the given arguments and an optional label. From hallon package.

Parameters

```
subfigure(
  ..args,
  outlined: bool,
  label: str,
  body
) -> content
```

outlined `bool`

Whether to outline this subfigure or not.

Default: `false`

label `str`

Unique label to reference this subfigure

Default: `none`

toc `content`

Custom-built `outline()`.

list-of-figures `content`

List of figures. Similar to LaTeX's `\listoffigures`.

list-of-tables `content`

List of tables. Similar to LaTeX's `\listoftables`.

theorem `content`

Theorem block.

proposition `content`

Proposition block.

lemma `content`

Lemma block.

remark `content`

Remark block.

proof `content`

Proof block.

Utilities

_is-page-empty

Check if a page is empty.

Parameters

```
_is-page-empty() -> bool
```

_empty-page

Adds an empty page between an odd page and the next. Used to check when to remove the header and place a raggiera in the bottom left corner.

Parameters

```
_empty-page() -> content
```

_nameref

It displays a reference using section name (instead of numbering). From Hallon 0.1.3

Parameters

```
_nameref(label) -> content
```

_raggiera-image

Inserts a raggiera, given a specified width.

Parameters

```
_raggiera-image(width: length) -> content
```

width `length`

Width of the raggiera.

banner

Draw a banner.

Parameters

```
banner(body: content) -> content
```

body `content`

Body of the banner.

tab-numbering

Alternative numbering: `"1.1." + h-space`.

Parameters

```
tab-numbering(
  h-space: length,
  ..n: arguments
) -> content
```

h-space `length`

Horizontal space.

Default: `0.7em`

..n `arguments`

Numbers.

chapter-numbering

Chapter numbering.

Parameters

```
chapter-numbering(..n: arguments) -> content
```

..n `arguments`

Numbers.

header-number

Numbering used in the theses header.

Parameters

```
header-number(..n: arguments) -> content
```

..n `arguments`

Numbers.

_sizes `dict`

LaTeX sizes to match original templates (https://tex.stackexchange.com/questions/24599/what-point-pt-font-size-are-large-etc)

_document-state `state`

The current state (title page, front-,main-,backmatter...)

_document-type `state`

The current type of document.

_localization `dict`

Localization dictionary. It could be deprecated since thesises are written exclusively in English.

bluepoli `color`

Signature PoliMi colour (■), used in headings and labels.

Figures styling

_subfigure-caption

It displays the caption of subfigures. Use "(a)" that will be colored based on `colored-headings` parameter.

Parameters

```
_subfigure-caption(
  colored-caption: bool,
  parent: figure,
  it
) -> content
```

colored-caption `bool`

Whether to apply bluepoli on captions

Default: `true`

parent `figure`

Parent figure of this subfigure

Default: `none`

_style-figures

Style-figures handles (optional heading-dependent) numbering of figures and subfigures. Can modify the caption of all figures.

Parameters

```
_style-figures(
  colored-caption: bool,
  heading-levels: int,
  body
) -> none
```

colored-caption `bool`

Whether to apply bluepoli on captions (Figure 1).

Default: `true`

heading-levels `int`

Heading level to account for in the numbering.

Default: `0`